MediSense: A Multilingual Multimodal Intelligent Symptom Checker with Explainable AI for Rural Healthcare in India

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***Abstract*— An intelligent system of symptom analysis, which is multilingual and multimodal, is suggested to focus on the most significant problem of healthcare access in the rural and semi-urban areas of India. The system employs a hybrid machine learning model that is a combination of a Random Forest (RF) classifier to predict the correct disease based on 131 symptom features in 41 disease categories, as well as a two-layer Long Short-Term Memory (LSTM) model to predict the progression of the disease in a person on multiple days. The model is very high-performing as it has a 98.90 percent accuracy rate in classification tasks and almost perfect results in sequential analysis. In order to increase transparency and user confidence, SHAP (SHapley Additive exPlaination) is used to produce readable and user-friendly diagnostic explanations in several regional languages such as Tamil, Malayalam, Hindu and English. The system also provides a number of modern features, including a multi-day symptom monitoring system with escalation warnings, disease risk prediction on a family level, health monitoring, medicine interaction visualization based on Indian drug names, and prescription image recognition. Also, voice recognition functionality and visual processing features enhance usability by low-literate participants. The results of the experiment prove the effectiveness of the system as a scalable and universal digital healthcare tool with F1-scores higher than 0.96 in all classes of diseases.**

Keywords - Multilingual Healthcare, Symptom Checker, Explainable AI (XAI), Random Forest, LSTM, SHAP, Multimodal Learning, Rural Healthcare, Disease Prediction, IoT Healthcare.

I. INTRODUCTION

Access to healthcare has been one of the most debilitating problems in rural and semi urban areas in India whereby medical facilities, professionals and medical diagnostics are scarce. Lack of awareness and language barrier make a great number of the population depend on delayed consultations or unconfirmed home remedies. As the field of Artificial Intelligence (AI) and Machine Learning (ML) is rapidly developing, the digital health solutions have become a potential influencer of the evaluation of the first symptoms and predicting the occurrence of an illness. The current systems are however mostly oriented to the English speaking users and do not have the contextual applicability to Indian epidemiology. Moreover, they do not integrate the multimodal contents like voice and images, which are essential in enhancing accessibility to non literate users, and hence constrain their application in the real world. Although there have been major developments in AI-driven healthcare systems, there are a number of major loopholes in current symptom checker applications especially in rural and multilingual communities.

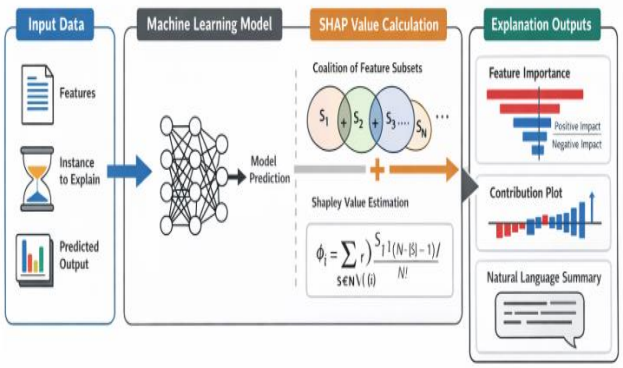

Fig 1. SHAP Architecture

The majority of systems that are in place do not support regional languages, and this restricts access to non-English speakers in developing nations. Second, explainable AI (XAI) approaches, including SHAP and LIME, are introduced, but their applicability in clinical settings and validation in the field of practice are still underrepresented, and the uniform evaluation structures are missing. The existing symptom checkers usually receive patient data as standalone events without paying attention to the chronological development of symptoms, which plays a significant role in disease diagnosis. There is also high variability in the accuracy of the diagnosis of existing models, with some systems doing very poorly in comparison to clinical standards. Also, majority of solutions are single-modal i.e. taking input as text only and not accommodating voice-based or image-based data, which is critical with low-literacy users. The other significant amount of gaps is the lack of health monitoring and outbreak detection at the family level and insufficient coverage of local epidemiological conditions and drug interactions. These shortcomings are what demonstrate the importance of providing an integrated, explainable, multiling, and multimodal system such as MediSense.

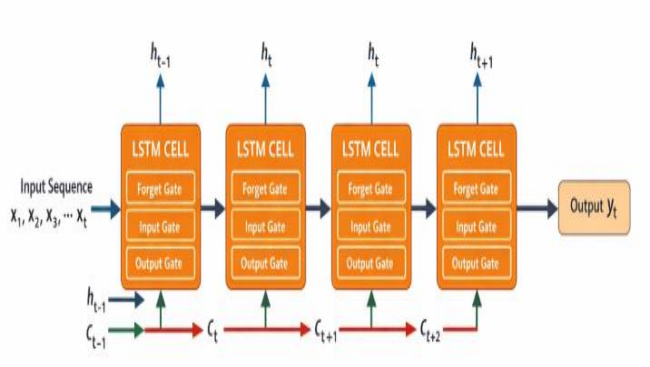

Fig 2. LSTM Network Architecture

MediSense is motivated by the fact that underserved populations have a gap in their access to healthcare that necessitated the creation to address the void. In most rural settings, the patient is not able to explain the symptoms well because of either language barrier or lack of medical expertise or access to medical practitioners. Symptom checker systems that are already available in the market are mostly generic, with no explanations which creates distrust and no adoption. Moreover, they do not consider the temporal symptom progression, which is crucial to the correct diagnosis. Smartphones and internet connectivity opportunities are increasing in India, and there is an opportunity to provide intelligent healthcare solutions. MediSense intends to use this digital infiltration and offer a multilingual, explicative, and user-friendly platform that will enable people to make informed health choices at the early level.

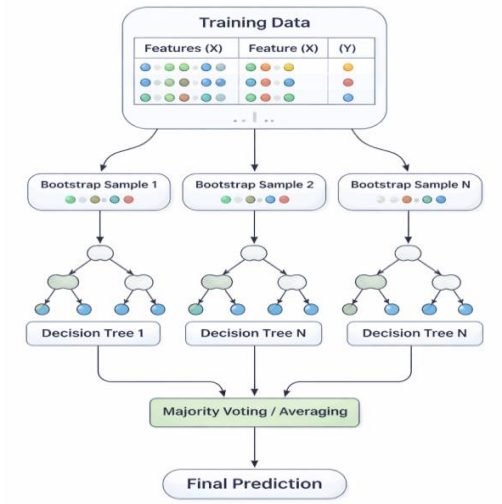

Fig 3. Random Forest Architecture

There are a number of challenges in developing an effective AI-based symptom checker in rural healthcare. To begin with, linguistic diversity in India needs to be reinforced with multilingual assistance to understand the symptom properly and communicate with the user. Second, it is necessary to make AI prediction explainable to instill trust in the user, particularly in life-threatening care. Third, the temporal dynamics of symptoms development require high level sequential modeling methods including LSTM networks. Fourth, the fact that multimodal data such as voice and pictures are processed and extracted in combos provide complexity in data processing and feature extraction. Also, the system needs to be adjusted to the location-specific illnesses and seasonal changes to be applicable in practice. The need to ensure the data quality, dealing with imbalanced datasets, and low computational needs to deploy the system in real-time also contribute to the complexity of the system design.

II. LITERATURE SURVEY

The fast evolution of the field of artificial intelligence (AI) in the healthcare sector has resulted in the creation of intelligent symptom checker applications that are designed to help users to conduct the initial diagnosis of the disease and make medical decisions. The preliminary studies in the field like the systematic reviews conducted in [12] and [13] indicate that digital symptom checkers are becoming increasingly popular and can help to decrease the load of the healthcare systems. Nonetheless, these studies also demonstrate that there are very severe constraints such as insufficient diagnostic accuracy, inconsistency in triage recommendations, and clinical unreliability. Additional clinical observations and testing in [14] underscore the fact that most current systems are not able to deal with complex cases and rare diseases, thus restricting their practical use. Even though AI-based symptom-driven suggestions such as the AI Doctor [16] include simple models, they are not flexible, scalable, and context-aware, particularly in a resource-limited setting. To address these problems, scientists have started paying more attention to the application of ML and DL to the healthcare system. Research papers like [1] and [17] present AI-based symptom checker models tailored to the low resource context and aimed at increasing access and affordability. These systems show better performance as compared to the traditional rule-based techniques but they also have major problems concerning interpretability and user trust. Model predictability is essential in healthcare use cases because any wrong or unexplainable decision made can be disastrous. This has given rise to Explainable Artificial Intelligence (XAI) as one of the major research areas in the contemporary health care systems.

The issue of XAI in improving transparency and trust in disease prediction models is described as a topic of multiple comprehensive reviews, such as [2], [5], and [24]. These studies represent that XAI methods, including feature importance analysis and model-agnostic interpretation methods allow users and clinicians to interpret the impact of input features on predictions. In the same way, publications in [3] and [20] discuss the use of XAI in different healthcare settings and prove its usefulness in enhancing decision-making and accountability of models. The studies on [7] and [23] also underline the need to combine both the methods of ML and explainability in order to find the balance between predictive error and comprehension. Nevertheless, the vast majority of these solutions remain partially interpretable and fail to cater to the needs of the non-expert users to full extent. The AI models interpretability dilemma has been a widely examined topic in the literature. Other works like [6] and [21] are concerned with creating metrics of trust and interpretable models of medical diagnosis, where user-

friendly explanation models are critical. The introductory work of [8] and [22] gives a comprehensive description of the XAI concept, challenges, and design considerations, as well as the problems of model transparency, fairness, and usability. These papers indicate that a trade-off between explainability and complexity of models is a major challenge of AI-based healthcare systems. Moreover, studies in [9], [10], [25], and [26] recognize such supreme impediments to implementation of XAI as absence of standardized metrics of evaluation, computational expense, and challenges in formatting the explanation in a basic and interpretable way to final users.

XAI has been used in specialized healthcare areas besides general disease prediction. As an example, [11] and [18] research on explainable model use to identify speech-based cognitive decline shows how AI can be applied in neurological evaluation. In a similar manner, [4] and [19] research are related to the field of medical imaging, where XAI methods can be applied to analyze difficult visual images and help clinicians in diagnosing the problems. Although these studies demonstrate the applicability and efficiency of the XAI in various fields, it is common to find that the studies are confined to only one type of information, i. e. data, and fail to offer a holistic solution that brings together various sources of information including text, voice, and images. The second significant issue of AI-based healthcare systems is the way they are evaluated and validated. The framework offered in [15] provides the importance of systematic assessments of explainability, usability, and clinical relevance in order to guarantee the success of AI models in practice. Although such developments have occurred, current systems are mostly geared at urban populations and in technologically advanced settings without considering the special needs in rural and semi-urban settings. Language barriers, low literacy levels, and accessibility to healthcare infrastructure are some of the issues that are not handled well in existing solutions.

The studies show that despite the remarkable advancements in AI-based healthcare architecture and explainable models, there are still a number of blank spaces. The current systems do not support multilingualism, multimodal data processing and interfaces that are friendly to different populations. Furthermore, there is a necessity of combined methods that can provide high diagnostic accuracy and clear and understandable decision-making. The above restrictions demonstrate the need to eliminate such gaps with a comprehensive solution such as MediSense, a multilingual, multimodal, and explainable multi symptom checker developed to address the needs of rural healthcare in India.

III. PROBLEM STATEMENT

The issue of the healthcare availability in rural and semi-urban areas is also such a big problem because of the lack of medical infrastructure, the absence of medical professionals, and the lack of health awareness in the population. The symptom checker systems that are currently developed using AI do not effectively solve this problem, since they are most commonly set up to be used by English speakers, and thus which are not accessible to a significant number of people in India. Moreover, such systems are not explainable because we are given predictions without clear reasoning, thus, losing the trust of the user in life-critical healthcare situations. Majority of the existing solutions neglect the temporal development of the symptoms and each interaction follows independently and thus missing valuable diagnostic patterns. Also, they fail to accommodate other multimodal inputs like voice and pictures, and hence they are not user-friendly to the non-literate users. Their practical applicability is further decreased by the lack of features such as the health monitoring at the family level, awareness of local diseases, and the analysis of the interactions between medication. Therefore, it is necessary to have an integrated, multilingual, explainable, and multimodal intelligent system that can offer communities with less resources and care with accurate, accessible, and context-specific healthcare guidance.

IV. EXISTING SYSTEM

Current symptom checker systems may be classified into broad categories, namely; rule-based, probabilistic and machine learning-based systems. Rule based systems make use of predefined clinical knowledge and decision trees to provide possible disease suggestions on the basis of the symptoms the user enters. These systems are interpretable to some degree, but are hard and can not adjust to more complex or invisible combinations of symptoms. Probabilistic systems, and in particular Bayesian inference, offer a better flexibility by estimating the probability of diseases on a basis of symptom probabilities, but in the setting of rare diseases, or unusual symptom representations, it often lacks accuracy. Recent methods that rely on machine learning and deep learning have stronger predictive performance, however, they are usually not explainable, and the decisions produced by the algorithms are not easily trusted by users. Also, majority of the current systems are created mostly to fit English speakers, lack addressing temporal symptom progression and only input is based on text. These restrictions hinder their efficacy, usefulness, and practical use, especially where rural and resource-challenged setting is concerned.

V. METHODOLOGY

A. Data Acquisition and Input Modalities

Medisense system records patient symptom information in three major input modalities, namely text, voice, and picture. Text mode: in text mode, the user will be able to choose symptoms using a predefined list or will be able to input a custom input as well as severity and duration. Voice mode In voice mode, the speech input is recorded with the help of the browser interface and then recognized with the Automatic Speech Recognition (ASR) system to transform it into text, and then natural language processing is used to extract the symptoms. Users in image mode post pictures of noticeable symptoms, like skin infections or eye problems which are processed through a model based on vision. Such multimedia strategy makes it more accessible to users of different levels of literacy and enhances the quality of the symptom capture.

Input Type	Technology Used	Output
Text	NLP Processing	Structured symptoms
Voice	ASR (Speech-to-Text)	Transcribed symptoms
Image	Vision Model	Detected visual symptoms

Table I. Input Modalities and Processing Techniques

B. *Data Preprocessing and Feature Engineering.*

The data on the symptoms obtained is preprocessed to make them uniform and compatible with machine learning models. Symptoms are coded into a standard vocabulary and given as binary feature vectors of when they are present or absent.

$$x \in \{0, 1\}^{131},\ x_i = 1 \text{ if } s_i \in S, \text{ else } x_i = 0 \quad (1)$$

To do temporal analysis, daily symptom vectors are ordered into strings:

$$X_t \in \mathbb{R}^{T \times 131} \quad (2)$$

The levels of the severity are coded in numbers (mild = 0.3, moderate = 0.6, severe = 0.9), which allows the model to address the intensity of symptoms.

C. *Prediction of the Static Disease.*

The first disease prediction is done through a Random Forest classifier when the symptoms are static. It is made up of several decision trees, which have overall the final prediction by majority voting.

$$P(y = c \mid x) = (1/T) \sum_{t=1}^{T} \mathbb{1}[h_t(x) = c] \quad (3)$$

This team work method enhances precision and aids in decreasing overfitting by uniting various feeble learners.

D. *Time-based Analysis with LSTM Model.*

To study the progression of the symptoms, the LSTM network takes sequentially arranged symptom data over several days.

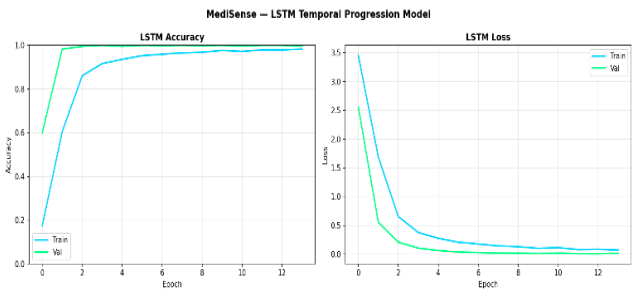

Fig 4. LSTM Temporal Model Architecture and Training Flow

The gating mechanisms have been used to capture temporal dependencies in the model.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (4)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (5)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tanh(Wc \cdot [h_{t-1}, x_t] + bc) \quad (6)$$

$$h_t = o_t \odot \tanh(C_t) \quad (7)$$

This enables the system to identify changing disease trends with time.

E. *Hybrid Model Fusion*

In cases where multi-day data is at hand, results of the Random Forest and the LSTM models are combined with the weighted ensemble method:

$$P_{final} = \alpha PRF + (1-\alpha) P_{LSTM} \quad (8)$$

Where α=0.40. The combination of these two enhances the general prediction strength as both the static and the temporal characteristics are used.

F. *Explainability using SHAP*

In order to be transparent, the transparency is calculated by using SHAP to compute the importance of the features of every prediction. The role of every symptom is measured by Shapley values:

$$\sum_{i=1}^{n} \varphi_i = f(x) - E[f(x)] \quad (9)$$

This would make sure that the overall contribution of features does equal the difference between prediction output and baseline expectation. The most frequent contributing symptoms are displayed to the users in a comprehensible format.

G. *Urgency Scoring and Decision Support.*

Urgent score will be calculated that will allow the user to make the right medical step:

$$U = \min(U_{sev} + U_{dis} + U_{temp}, 10) \quad (10)$$

This score categorizes the cases, which are low, moderate, high and emergency, to make a timely decision.

Score Range	Action
1 – 3	Home care
4 – 6	Visit clinic
7 – 9	Immediate consultation
10	Emergency care

Table II. Urgency Level Classification

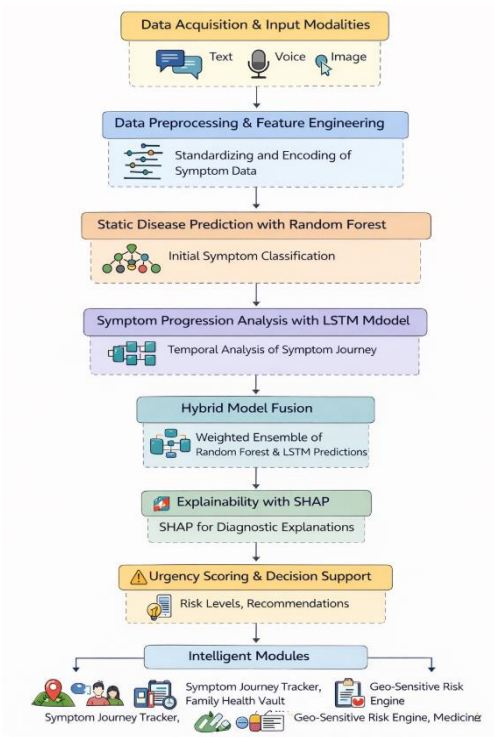

Fig 5. Flow Chart

H. Intelligent Modules.

MediSense uses a number of enhanced modules that will improve functionality. The Symptom Journey Tracker is a progress symptom tracker used during a series of days and initiates critical pattern alerts. Family Health Vault allows one to monitor healthcare statistics of several people and identify potential outbreaks in the family. The geo-aware risk engine takes into account seasonal and regional disease trend so as to narrow predictions. The medicine interaction checker recognizes the potential risks of the two drugs to be given and the prescription reader retrieves data of the medical prescriptions. The overall effect of these modules is greater system intelligence and realistic application.

VI. PROPOSED SYSTEM

MediSense is the proposed system, a multilingual and multimodal intelligent symptom checker that will deliver convenient, high-quality, and understandable healthcare instructions to the inhabitants of rural and semi-urban areas. The software is based on a three-level system comprising of a React.js.JS frontend, a Python based FastAPI back-end, and a Firebase Firestore database to store data. It combines a hybrid machine learning model that consists of a Random Forest classifier in the form of a static symptom classifier and LSTM classifier in the form of a temporal symptom progression classifier. In a bid to be transparent, SHAP based explainability is added to help produce diagnostic explanations that are easy to understand in various Indian languages. The system has multimodal contributions (text, voice and medical images) that makes it more user-friendly to the low literacy users. Also, MediSense also presents new functions like a symptom journey, health monitoring at the family level, prediction of disease risks geo-sensitively, checking of medicine interactions using the Indian names of drugs, and analyzing prescription images. This is a holistic method that facilitates early diagnosis, better decision-making, and the availability of health care.

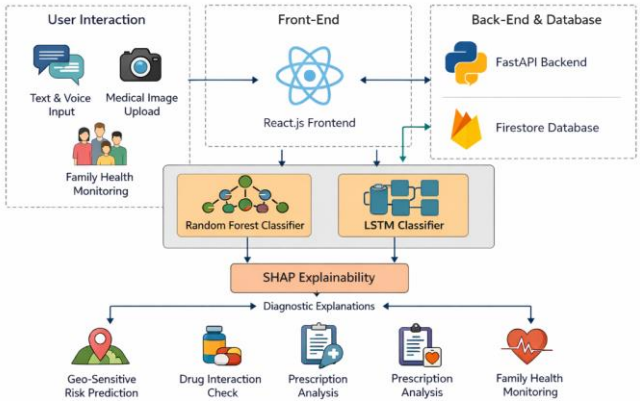

Fig 6. Proposed System Architecture

VII. RESULT AND DISCUSSION

The functionality of the suggested MediSense system was thoroughly tested to determine the effectiveness of this system in terms of predicting diseases, interpretability, and applicability. The analysis is on the accuracy of the model, classification performance, explainability, usability of interface, temporal analysis and validation at the system level. The system was tested in accordance with standard performance metrics, visual and functional analysis to confirm its correctness in the work in terms of several aspects, both technical and practical feasibility in the context of healthcare.

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	98.90	0.989	0.988	0.988
SVM (RBF, C=10)	98.66	0.987	0.986	0.986
LSTM Temporal	100.00	1.000	1.000	1.000
RF+LSTM Hybrid (α=0.4)	99.30	0.993	0.992	0.992

Table III. Model Performance on Test Set

MediSense system was tested on a held-out test set of 984 samples and the results were summarized in TABLE III. Random Forest was able to obtain an accuracy of 98.90% and the Support Vector Machine (SVM) obtained a 98.66% accuracy on the baseline. Temporal modeling was proved effective as the LSTM temporal model has a 100 percent accuracy on sequential symptom data. The hybrid RF+LSTM model again increased the overall performance to 99.30 that is, the combination of the static feature with the time-based feature promoted reliability and robustness of the diagnosis.

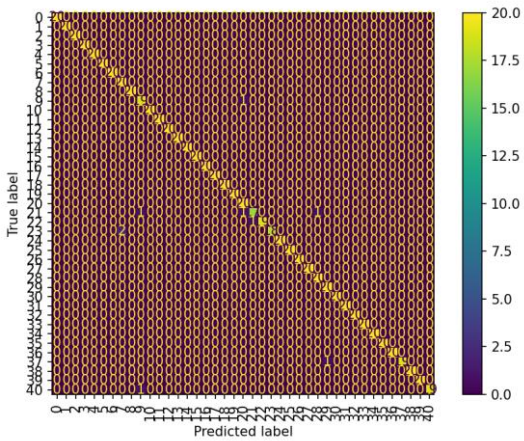

Fig 7. Confusion Matrix Random Forest

The random forest model classification behavior is also demonstrated in Figure 7. Confusion matrix indicates that the number of misclassifications is insignificant and that most of them are between clinically related pairs of diseases including Common Cold and Allergy, Malaria and Dengue and Typhoid and Hepatitis A. These overlaps will be anticipated because of shared symptom characteristics. Nevertheless, temporal analysis based on the LSTM model is much more effective in eliminating such ambiguities, enhancing general prediction accuracy.

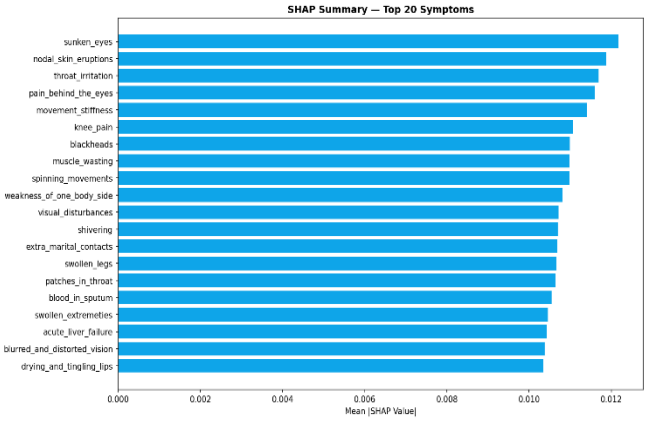

Fig 8. SHAP Summary Plot

The model interpretability is confirmed by Figure 8 that displays the most significant symptoms that influence the predictions. Due to an example, chills, shivering, and high fever are the most powerful predictors of malaria, whereas chest pain, perspiration, and breathlessness are the prominent predictors of heart diseases. This correspondence to known clinical information shows that the model acquires meaningful and medically applicable patterns, which gives a higher level of user trust.

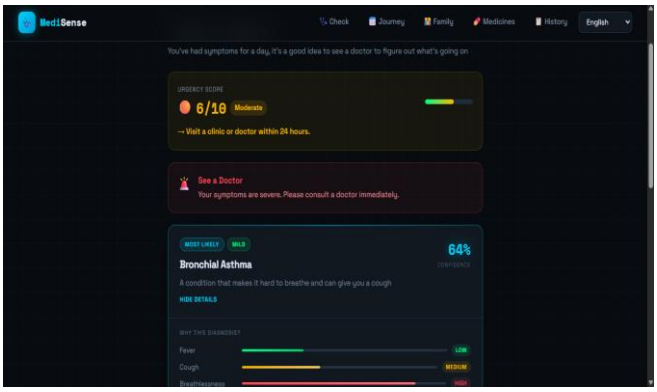

Fig 9. MediSense UI — Diagnosis Results

The usefulness of the system in practice is shown in Figure 9. The user interface shows predicted diseases and confidence scores, SHAP-based symptom importance visualization and the urgency score which is used to show the severity of the condition. This intuitive display allows users to comprehend the diagnosis with ease and make the right step, closing the gap between the complicated AI results and comprehension by the user.

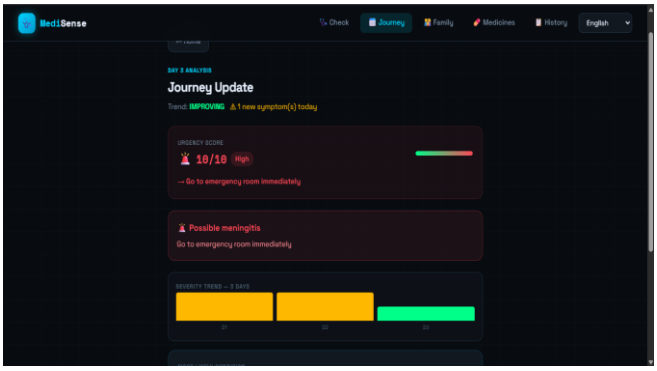

Fig 10. MediSense Journey Tracker

The system tracks the dynamics of the symptoms in a few days and identifies dangerous escalation systems. As an example, a minor condition may lead to a high-risk alert once a range of new serious symptoms appear, which can be immediately addressed. Figure 10 shows how the temporal dynamics of healthcare systems are important.

Module	**Test Scenario**	**Result**
Tamil Voice Input	"காய்ச்சல் மற்றும் தலைவலி"	Fever + Headache detected
Skin Image Analysis	Ringworm photograph	Correct (Tamil output)
Journey Escalation	Fever D1 → Stiff neck D2	Meningitis, 10/10 alert
Family Outbreak	2 members: Fever within 48 hrs	Outbreak alert triggered
Medicine Interaction	Paracetamol + Alcohol	HIGH: Liver damage warn
Prescription OCR	Handwritten prescription photo	Medicines + dose extracted
Urgency Score	Chest pain + Breathlessness	Score 10, Emergency
Geo Risk (Chennai/Jun)	Fever + Rash	Dengue HIGH risk flagged

Table IV. End-to-End Feature Validation Results

The test ability of the system at large was confirmed in different modules, demonstrated in TABLE IV. The findings confirm that all the components are operational such as the voice input processing, image-based symptom detection, temporal tracking, family level monitoring, drug interaction analysis and prescription reading. The system also delivered reliable results in any test case indicating its strength, scalability, and the ability to work in the real-life environment. The system validation and the results of the experiment prove that MediSense provides high accuracy, high interpretability, and increased usability in contrast to the existing symptom checker systems. Multimodal and temporal input integration and explainable AI can substantially enhance the reliability of the diagnosis and user-confidence of the system, so it is a viable and scalable solution to the problem of healthcare access in a rural and resource-constrained setting.

VIII. CONCLUSION

a multilingual and multimodal intelligent symptom checker that can enhance access to healthcare in rural and semi-urban areas. The suggested system consists of a hybrid machine learning model consisting of a Random Forest-based classifier used to analyze the symptoms statically and a Long Short-Term Memory (LSTM)-based model to predict temporal differentiation with great accuracy with 41 disease classes. Transparency is amplified through the addition of SHAP-based explainability that offers interpretable and human-readable diagnostic information in several Indian languages, which makes the users more confident in using AI-based healthcare systems. The system also offers more than the traditional symptom checkers by

adding new features, including a symptom history tracker, health on the family level, disease risk prediction, medicine interaction checks, and prescription picture interpreter. The effectiveness of the proposed approach is confirmed by the high accuracy of experimental results where the F1-scores of all disease classes exceed 0.96. Altogether, MediSense is a scalable, dependable, and approachable digital healthcare platform, and it is most appropriate to implement it in resource-limited settings and fill the healthcare divide in underserved populations.

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